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From Theory to Practice: An Empirical Audit of P3 Candlestick Patterns Under Realistic Transaction Costs and Market Regimes

Methodological Framework for Rigorous Backtesting

The empirical validation of trading protocols demands a methodological framework that transcends simplistic pattern recognition and addresses the multifaceted nature of modern financial markets. The research goal specifies a stringent set of requirements designed to ensure that any identified edge is robust, statistically significant, and deployable in live trading conditions. This involves moving beyond raw, unfiltered pattern statistics—which are often misleading—to an analysis conditioned upon a comprehensive stack of filters and instrument-specific calibrations [140]. The core principle guiding this methodology is that the edge is not inherent in the pattern itself but is created through the convergence of multiple, context-aware criteria that isolate high-probability setups from market noise [25]. This section deconstructs the essential components of this rigorous framework, focusing on the critical role of transaction cost modeling, statistical significance thresholds, and the imperative to validate performance in structurally distinct market regimes, particularly those emerging post-2020 [21] [23].

A foundational element of the proposed methodology is the explicit prioritization of institutional or proprietary-style backtests over purely academic studies [78]. While academic papers offer valuable theoretical insights, they frequently rely on idealized assumptions such as zero transaction costs, continuous data without gaps, and perfect order execution [51]. These simplifications can create a veneer of profitability that evaporates upon real-world implementation. In contrast, institutional backtests incorporate granular details of market microstructure, including bid-ask spreads, slippage, partial fills, and the impact of session liquidity [45] [120]. For instance, when testing OTC instruments like Forex pairs or CFDs, entry prices should be modeled using the ask price and exits using the bid price to account for the spread, which directly erodes potential profit [27]. Slippage, the difference between the expected price of a trade and the price at which the trade is actually executed, must also be quantified. This can be

modeled based on historical tick volume proxies or multiples of the Average True Range (ATR) to reflect increased execution uncertainty during volatile periods [95]. Furthermore, certain instruments, such as energy commodity CFDs, are subject to rollover costs when positions are held past their expiration date, a tangible drag on performance that must be factored into expectancy calculations [45]. The preference for backtests that explicitly model these frictions ensures that the resulting win rates and expectancies are realistic and reflective of a trader's true potential returns after costs [95].

The second pillar of the framework is the strict adherence to statistical significance. The directive mandates a minimum sample size of N=30 for any valid conclusion [78]. This threshold serves as a crucial safeguard against drawing erroneous conclusions from small, anomalous datasets—a common pitfall in trading strategy development known as overfitting. A pattern might appear highly profitable in a limited historical window, but its performance may not be sustainable. By requiring a larger sample base, the methodology promotes robustness and generalizability. When a test case fails to meet this N=30 threshold, the result must be flagged with a low confidence level and accompanied by the 95% confidence interval [78]. This approach provides a transparent measure of uncertainty, acknowledging that conclusions drawn from smaller samples are less reliable. It prevents the smoothing out of critical failures that might occur in low-volume environments by forcing a separate evaluation for each instrument class, thereby preventing a successful pattern on one asset from masking a fatal flaw on another [45]. The focus on per-instrument evaluation is paramount, as the unique characteristics of each asset class—including volatility, liquidity profiles, and trading hours—necessitate distinct calibration parameters [26] [82] [119].

Finally, the framework must be dynamic enough to adapt to changing market regimes. The user's directive to give 70% weight to post-2020 data and to stratify results by VIX volatility regimes is a direct acknowledgment of the Adaptive Market Hypothesis (AMH) [21]. According to this hypothesis, market efficiency is not constant but shifts over time in response to technological change, regulation, and participant behavior [21]. The proliferation of high-frequency trading (HFT), algorithmic liquidity pools, and fragmented trading sessions since 2020 has structurally altered market dynamics, potentially degrading traditional technical patterns that were effective in earlier eras [21] [23]. Therefore, a pattern's long-term geometric integrity does not guarantee its future profitability [140]. The analysis must dissect performance across Low (VIX < 16), Normal ($16 \leq$ VIX ≤ 25), and High (VIX > 25) volatility regimes [57] [91]. A pattern that demonstrates a 55-65% win rate claim in a stable, low-volatility environment but collapses to below 50% in a turbulent, high-volatility setting would be considered

unreliable for live deployment, regardless of its overall historical performance [21]. This regime-aware approach ensures that the protocol's viability is tested under the precise conditions where it would be deployed, providing a more honest assessment of its practical utility. The requirement to use pre-2020 data primarily for verifying pattern geometry further reinforces this focus on contemporary market realities [21]. This comprehensive methodology, combining realistic cost modeling, statistical rigor, and adaptive regime analysis, forms the necessary foundation for a credible empirical validation of the Protocol 3 Tier 2 patterns.

Performance Analysis of Single-Candle Reversal Patterns

Single-candle reversal patterns, namely the Hammer, Shooting Star, and Engulfing patterns, represent some of the most fundamental formations in technical analysis, designed to signal a potential shift in market sentiment after a prevailing trend [88] [140]. Their mechanical definitions are straightforward, making them attractive for systematic trading protocols. However, their reliability hinges critically on contextual factors such as trend alignment, momentum exhaustion, and confirmation from subsequent price action. The research goal mandates a rigorous evaluation of these patterns under a strict 8-filter convergence stack, which includes trend alignment via the EMA(200), momentum confirmation via RSI, and volatility filtering via ATR, among others [87] [140]. This section analyzes the empirical performance of these patterns, synthesizing available information to assess whether they can sustain a positive expectancy after accounting for realistic transaction costs and microstructural nuances.

The Hammer and its bearish counterpart, the Shooting Star, are characterized by a small body located at the top or bottom of the candle range, respectively, and a long lower or upper wick, indicating that aggressive buying or selling pressure was ultimately rejected by the opposing force [88] [140]. The protocol's specific mechanical definition requires a lower wick for a Hammer to be at least twice the size of its body, with an upper wick no more than a quarter of the body size, and the close occurring in the upper third of the range [140]. This stringent definition aims to filter out weak or indecisive wicks. The primary failure mode for these patterns often relates to false confirmation. A stop-order placed beyond the extreme of the hammer candle may be triggered by a minor price movement that fails to lead to a sustained reversal, resulting in a loss as the price quickly retraces [27]. The presence of a mandatory confirmation filter, requiring the next candle to

close beyond the pattern's extreme, is designed to mitigate this exact failure mode [140]. However, in choppy, high-volatility markets, even confirmed signals can be premature if they occur at a point of exhaustion rather than capitulation. The ADX regime gate, which requires ADX(14) ⩾ 25, acts as a crucial filter here, ensuring that the initial setup occurs within a strong trend that is ripe for reversal, thus increasing the probability of a meaningful move [84]. Without this filter, a hammer appearing in a consolidation phase may simply mark a continuation of sideways price action.

The Bullish and Bearish Engulfing patterns consist of two candles, where the body of the second candle completely engulfs the body of the first [140]. The protocol adds specific requirements to enhance their reliability: the engulfing candle must form when RSI(14) is below 35 (for bullish) or above 65 (for bearish), and trading volume must exceed twice the 20-period simple moving average of volume [140]. The RSI filter ensures the pattern appears at a point of potential oversold/overbought conditions, while the volume filter confirms that the reversal is supported by significant participation. The primary challenge with engulfing patterns lies in their susceptibility to being part of a broader, stronger trend. A single engulfing candle may be nothing more than a temporary pullback or a failed reversal attempt. The multi-filter stack is designed to address this. The EMA(200) trend alignment filter ensures the trade direction is aligned with the master trend, positioning the engulfing pattern as a counter-trend rally or dip within a larger directional move [87]. The volatility filter (pattern range > 1.0× ATR(14)) ensures the pattern represents a meaningful price move relative to recent volatility, not just a minor fluctuation [140]. Despite these enhancements, the main failure mode remains premature entries. Traders might enter too early, before the confirmation candle closes beyond the pattern's extreme, leading to stops being hit by short-term noise. The protocol's rule for placing a stop order one pip beyond the confirmation extreme is intended to enforce patience and wait for concrete evidence of a reversal before committing capital [140].

For precious metals like Gold (XAUUSD) and Silver (XAGUSD), the standard parameters require instrument-specific calibrations. For Gold, the volatility filter is tightened to a range greater than 1.5× ATR(14), and traders are advised to avoid entering trades around LBMA fixing times (10:30 and 15:00 London time) [26]. These adjustments recognize Gold's sensitivity to physical market events and its tendency to exhibit higher volatility compared to Forex majors. For Silver, the calibration is even more pronounced: the ATR filter is raised to 1.75× ATR, the RSI thresholds are widened to <20 and >80, and the confirmation requirement is made stricter, demanding the price clears the pattern extreme by at least 1× spread + 0.1 ATR [140]. These changes reflect Silver's reputation for high volatility and frequent whipsaws. The wider RSI bands account for the fact that momentum can remain extremely strong for longer periods before reversing.

The tighter confirmation requirement is a direct response to Silver's propensity for sharp, erratic moves that can easily trigger a stop-loss without a genuine reversal. These specific calibrations underscore the necessity of a per-instrument evaluation, as applying standard Forex rules to Silver could lead to a high frequency of false signals and negative expectancy [90] [119]. The lack of publicly available institutional backtests specifically targeting these calibrated patterns makes it difficult to assign definitive win rates and expectancy values, highlighting a gap in the available empirical literature. Any conclusion would be speculative without a dedicated, granular backtest adhering strictly to the provided protocol.

Pattern	Instrument	Filtered WR%	Net Expectancy (R)	Sample Size	Source/ Citation	Verdict (Keep/ Downgrade/ Remove)	Primary Failure Mode
Hammer	Forex Majors	Not Available	Not Available	Not Available	Not Available	Keep	Wick Overshoot / False Confirmation
Shooting Star	Forex Majors	Not Available	Not Available	Not Available	Not Available	Keep	Wick Overshoot / False Confirmation
Bullish Engulfing	Forex Majors	Not Available	Not Available	Not Available	Not Available	Keep	Premature Entry / Counter-Trend Mismatch
Bearish Engulfing	Forex Majors	Not Available	Not Available	Not Available	Not Available	Keep	Premature Entry / Counter-Trend Mismatch
Hammer	Gold (XAUUSD)	Not Available	Not Available	Not Available	Not Available	Keep	Whipsaw / LBMA Fix Event Noise
Shooting Star	Gold (XAUUSD)	Not Available	Not Available	Not Available	Not Available	Keep	Whipsaw / LBMA Fix Event Noise
Bullish Engulfing	Gold (XAUUSD)	Not Available	Not Available	Not Available	Not Available	Keep	Premature Entry / Lower Volatility Sensitivity
Bearish Engulfing	Gold (XAUUSD)	Not Available	Not Available	Not Available	Not Available	Keep	Premature Entry / Lower Volatility Sensitivity
Hammer	Silver (XAGUSD)	Not Available	Not Available	Not Available	Not Available	Downgrade	Extreme Whipsaws / Tight Stop Losses
Shooting Star	Silver (XAGUSD)	Not Available	Not Available	Not Available	Not Available	Downgrade	Extreme Whipsaws / Tight Stop Losses
Bullish Engulfing	Silver (XAGUSD)	Not Available	Not Available	Not Available	Not Available	Downgrade	Extreme Whipsaws / Tight Stop Losses
Bearish Engulfing	Silver (XAGUSD)	Not Available	Not Available	Not Available	Not Available	Downgrade	Extreme Whipsaws / Tight Stop Losses

Performance Analysis of Multi-Candle Reversal Patterns

Multi-candle reversal patterns, such as the Three Inside Up/Down and the Morning/Evening Star, are generally considered more robust than their single-candle counterparts because they depict a more extended process of trend exhaustion and capitulation [80] [140]. These patterns unfold over three days, illustrating a clear narrative: a dominant trend pauses, followed by a powerful reversal candle that overwhelms the countertrend. The protocol's mechanical definitions for these patterns are strict, requiring precise relationships between the bodies and ranges of the constituent candles [140]. For example, the Three Inside pattern requires the second day's body to be fully contained within the first day's body, and the third day's close must exceed the high of the first day [140]. Similarly, the Evening Star pattern necessitates a final closing price that is more than 50% into the body of the first candle, along with a confirming fourth candle [140]. The analysis of these patterns' empirical validity requires evaluating whether this added complexity translates into a higher win rate and net expectancy after applying the full suite of eight filters and accounting for realistic trading frictions.

The Morning Star and Evening Star patterns are potent reversal signals consisting of three candles: a large candle in the direction of the prior trend, a small indecision candle (like a doji or spinning top) representing a pause, and a large candle in the opposite direction signaling the reversal [22] [80]. The protocol's specification that the third candle's close must be more than 50% into the body of the first candle is a key strength. This ensures that the reversal is not just a minor pullback but a substantial penetration of the previous trend's momentum [140]. The primary failure mode for star patterns is often a premature reversal that quickly gets absorbed back into the larger trend. The multi-filter stack is designed to combat this. The ADX regime gate (ADX $\geqslant$ 25) is particularly crucial, as it ensures the pattern emerges from a strong, established trend, making a reversal more probable [84]. The session timing filter (London or NY session) targets periods of maximum liquidity, which increases the likelihood of a confirmation candle forming cleanly and decisively [140]. However, these patterns can still fail due to "stop-hunting" behavior prevalent in HFT-dominated markets. Algorithms may push price slightly beyond the pattern's extreme to trigger stop-loss orders before reversing, causing a confirmed signal to turn into a losing trade. The requirement for a 1:2 risk-reward ratio and scaling out at +1R and +2R helps manage this risk, but the underlying failure mode of a deceptive confirmation remains a concern [140].

The Three Inside Up/Down pattern consists of a long candle in the direction of the trend, a smaller candle of the opposite color whose entire body is contained within the range of

the first candle, and a third candle that closes beyond the high of the first candle [140]. This pattern suggests that sellers who attempted to take control of the market were overwhelmed by buyers, leading to a resumption of the original trend. The protocol's strict mechanical definition helps eliminate ambiguous formations. The main weakness of this pattern is its reliance on the third candle's confirmation. If the market enters a period of low volatility or consolidation after the second candle, the third candle may fail to form, leaving the trade idea invalid. The volatility filter (pattern range > 1.0× ATR) helps screen for setups with sufficient momentum to have a chance of producing a decisive third candle [140]. Another potential failure mode is the misinterpretation of the pattern within a complex market structure. For instance, if the pattern forms at a major support/resistance level that is simultaneously acting as a magnet for stop-loss orders, the resulting price action can be chaotic and unpredictable. The EMA(200) trend alignment filter helps by ensuring the pattern is interpreted correctly within the context of the dominant macro trend [87]. For example, a Three Inside Up pattern appearing after a prolonged downtrend near the EMA(200) is a much stronger buy signal than one appearing mid-trend [87].

Similar to single-candle patterns, multi-candle patterns require instrument-specific calibrations. For Gold, the recommendation is to prefer the Evening Star pattern over the Bearish Engulfing pattern [26]. This suggests that the multi-day formation of the star pattern may be better suited to Gold's typical price action and volatility profile than the abrupt two-day move of the engulfing pattern. For Index CFDs, there are specific rules related to the VIX and NYMEX trading hours. During periods of very high volatility (VIX > 30), filters are widened and position sizes are halved, suggesting that the standard pattern mechanics may break down in such chaotic environments [140]. The instruction to skip the first 30 minutes of the NYSE open and to shorten the time stop for shorts indicates that index prices are particularly prone to erratic opening behavior and rapid reversals, which can invalidate otherwise sound pattern setups [140]. The absence of large-scale, public backtests that detail the performance of these specific, filtered multi-candle patterns means that their empirical win rates and expectancy remain largely unvalidated. Their inclusion in the protocol is based on a combination of classical technical analysis theory and logical adjustments for different market microstructures, but a rigorous, quantitative validation under the specified conditions is required to confirm their efficacy [80] [140]. Without this, they must be considered as having a moderate to high degree of uncertainty until proven otherwise.

Pattern	Instrument	Filtered WR%	Net Expectancy (R)	Sample Size	Source/ Citation	Verdict (Keep/ Downgrade/ Remove)	Primary Failure Mode
Morning Star	Forex Majors	Not Available	Not Available	Not Available	Not Available	Keep	Deceptive Confirmation / Stop-Hunt
Evening Star	Forex Majors	Not Available	Not Available	Not Available	Not Available	Keep	Deceptive Confirmation / Stop-Hunt
Three Inside Up	Forex Majors	Not Available	Not Available	Not Available	Not Available	Keep	Lack of Third-Candle Confirmation
Three Inside Down	Forex Majors	Not Available	Not Available	Not Available	Not Available	Keep	Lack of Third-Candle Confirmation
Morning Star	Gold (XAUUSD)	Not Available	Not Available	Not Available	Not Available	Keep	Delayed Reversal / Lower Volatility
Evening Star	Gold (XAUUSD)	Not Available	Not Available	Not Available	Not Available	Keep	Delayed Reversal / Lower Volatility
Three Inside Up	Gold (XAUUSD)	Not Available	Not Available	Not Available	Not Available	Keep	Lack of Third-Candle Confirmation
Three Inside Down	Gold (XAUUSD)	Not Available	Not Available	Not Available	Not Available	Keep	Lack of Third-Candle Confirmation
Morning Star	Indices	Not Available	Not Available	Not Available	Not Available	Keep	Opening Whipsaws / Rapid Reversals
Evening Star	Indices	Not Available	Not Available	Not Available	Not Available	Keep	Opening Whipsaws / Rapid Reversals
Three Inside Up	Indices	Not Available	Not Available	Not Available	Not Available	Keep	Opening Whipsaws / Rapid Reversals
Three Inside Down	Indices	Not Available	Not Available	Not Available	Not Available	Keep	Opening Whipsaws / Rapid Reversals

Instrument-Specific Calibration and Microstructure Impact

The empirical validation of any trading protocol is incomplete without acknowledging the profound impact of instrument-specific microstructure. The research directive mandates a strict separation of analysis by asset class—Forex, Gold, Silver, Oil, and Indices—and the application of unique calibration rules for each [140]. This requirement stems from the

understanding that no single set of parameters can optimally capture the behavior of assets with vastly different characteristics, such as volatility profiles, liquidity patterns, trading hours, and fundamental drivers [29] [72]. Aggregating results across these classes would mathematically obscure critical, asset-class-specific failure modes, rendering any derived conclusions practically useless for deployment [140]. This section examines how the specified calibrations for each instrument class are designed to adapt the universal filter stack to the unique dynamics of that market, and synthesizes the likely impact of these differences on the performance of the Tier 2 patterns.

For Forex majors, the protocol relies on the standard, unadjusted parameters. This implies that the baseline filter settings (e.g., ATR multiplier of 1.0, RSI thresholds of 35/65) are considered appropriate for the relatively stable and liquid nature of major currency pairs [140]. The primary challenges in the Forex market relate to session overlaps and news-driven volatility. The session timing filter, restricting entries to London (08:00–16:00 GMT) or New York (13:00–21:00 GMT) hours, is designed to capitalize on peak liquidity and reduce the risk of trading in thin, illiquid markets where spreads widen and slippage increases [27] [140]. The exception for JPY pairs in the Asian session highlights the unique liquidity characteristics of that currency pair, allowing for a controlled exposure during its most active hours [140]. The effectiveness of the filter stack on Forex will depend heavily on its ability to navigate the predictable yet sometimes chaotic volatility of session opens and closes, as well as major economic data releases, which are handled by the news blackout filter [140].

Precious metals, specifically Gold (XAUUSD) and Silver (XAGUSD), require significantly more tailored calibrations due to their higher volatility and sensitivity to different market drivers. For Gold, the volatility filter is increased to a pattern range greater than 1.5× ATR, acknowledging that smaller movements are less likely to signify a meaningful reversal in its price action [140]. The prohibition on entering trades around LBMA fixing times (10:30 and 15:00 London time) is a critical adjustment. These are moments of high concentration of physical gold settlement activity, which can induce artificial price spikes and make technical patterns unreliable [26]. The preference for the Evening Star pattern over the Bearish Engulfing pattern suggests that Gold's reversals tend to be more protracted and less abrupt, fitting the three-candle narrative of the star pattern better [26]. For Silver, the calibrations are even more aggressive. The volatility filter is raised to 1.75× ATR, and the RSI thresholds are widened to <20 and >80, reflecting Silver's reputation for violent, fast-moving trends and deep, persistent pullbacks [140]. The recommendation to prefer single-candle sweeps like the Hammer and Shooting Star over multi-candle patterns may be due to Silver's tendency to reverse so quickly that multi-day patterns often fail to complete. The stricter confirmation requirement (clearing the

extreme by ⩾ 1× spread + 0.1 ATR) is a defensive measure against the extreme whipsaw risk inherent in trading Silver [140]. These calibrations indicate that while the same conceptual patterns apply, their mechanical expression and success rates are fundamentally different in the precious metals markets compared to Forex.

Energy commodities like Oil (USOIL/WTI) introduce a completely different layer of complexity driven by fundamental supply-and-demand shocks and scheduled news events. The protocol's calibrations for Oil are heavily influenced by these factors. The EIA and API inventory report blackouts (90m and 45m post-release, respectively) are essential, as these reports can cause massive, unpredictable price swings that render any technical setup obsolete [140]. Trading exclusively within NYMEX hours (09:00–14:30 ET) aligns with the primary trading session for these futures contracts [140]. The directive to suspend trading during CFD roll windows is a crucial microstructural consideration; rolling a futures contract involves closing one expiring contract and opening another with a later expiration, which can cause significant price discontinuities and slippage [45]. The statement that fundamentals override technicals is perhaps the most important rule for Oil, emphasizing that in this market, chart patterns are secondary to understanding global crude inventories, geopolitical tensions, and production levels [82] [83]. The empirical validation of candlestick patterns in Oil would need to demonstrate exceptional resilience to these powerful fundamental forces.

Index CFDs present another unique challenge, primarily due to their strong correlation with market-wide volatility, measured by the VIX. The protocol's VIX-based tiered system is a sophisticated adaptation to this reality [140]. In low-VIX environments (VIX < 16), the rule to skip all shorts acknowledges that indices rarely experience sustained bear markets during calm periods, making short signals highly suspect [140]. As volatility normalizes (VIX 20-30), standard parameters apply [140]. In high-VIX environments (VIX > 30), the filters are intentionally widened and position sizes are reduced, recognizing that mean-reversion strategies can break down and trends can become explosive [140]. The restriction on skipping the first 30 minutes of the NYSE open is a response to the "opening gap" phenomenon, where indices often gapped sharply in one direction before establishing a trend. The shortened time stop for short trades reflects the documented tendency of indices to recover rapidly from intraday dips, especially during periods of fear and uncertainty [69]. The empirical validation of this protocol for indices would be a strong test of its ability to adapt to shifting market sentiment and volatility regimes, which are central to index performance.

Instrument Class	Key Calibrations & Rules	Primary Microstructural Drivers	Likely Failure Modes
Forex Majors	Standard filters. London/NY session only. JPY exception in Asian session.	Liquidity cycles, news event volatility.	Low liquidity slippage, session open chaos.
Gold (XAUUSD)	Vol filter: >1.5× ATR. Avoid LBMA fixes (10:30/15:00). Prefer Evening Star.	Safe-haven flows, US dollar strength, central bank activity.	LBMA fix-induced whipsaws, delayed reversals.
Silver (XAGUSD)	Vol filter: >1.75× ATR. RSI: <20/>80. Stricter confirmation. Prefer single-candle sweeps.	Industrial demand, speculative retail flows, inverse correlation with USD.	Extreme whipsaws, tight stop losses being hit, pattern completion failure.
Oil (USOIL/WTI)	EIA/API blackouts. NYMEX hours only. Suspend during CFD rolls. Fundamentals override.	Supply/demand shocks, OPEC+ decisions, geopolitical risk, inventory data.	News event volatility, contract rollover slippage, trend override.
Index CFDs	VIX-tiered filters: Skip shorts if VIX<16. Widen filters/higher VIX. Skip first 30m open. Short time stop: 5 bars.	Broad market sentiment, interest rate expectations, economic data, corporate earnings.	Opening gap failure, rapid mean-reversion, explosive trends breaking filters.

Regime Dependency and Post-2020 Market Adaptation

The assertion that candlestick patterns possess timeless, universal efficacy is increasingly challenged by the realities of modern financial markets. The user's directive to prioritize post-2020 data and to stratify analysis by VIX-regulated volatility regimes is a direct response to this challenge, grounded in the Adaptive Market Hypothesis (AMH) [21]. According to AMH, market conditions are not static; they evolve due to technological advancements, regulatory changes, and shifts in participant behavior [21]. Since 2020, the market landscape has been reshaped by several key factors: the dominance of high-frequency trading (HFT) algorithms, the fragmentation of liquidity across multiple electronic venues, and heightened sensitivity to macroeconomic news, leading to extreme volatility episodes [21] [23]. These structural changes have the potential to degrade or even invert the historical probabilities associated with classic technical patterns. Therefore, a pattern's geometric shape, while enduring, may no longer reliably predict future price action in the same way it did in the more linear, trend-driven markets of the past [140].

The VIX, or CBOE Volatility Index, serves as a proxy for market-wide fear and expectations of future volatility, making it an ideal tool for stratifying market regimes [2] [91]. The research plan divides performance into three buckets: Low Volatility (VIX < 16), Normal (16 ≤ VIX ≤ 25), and High Volatility (VIX > 25) [140]. A pattern's success is contingent on its ability to perform across these diverse environments. For instance, a Hammer pattern forming in a Low Volatility regime, where markets are in a steady

uptrend, is likely to be a reliable buy signal. The market structure is supportive, and the probability of a reversal is low, meaning the pattern is more likely to be a minor pullback. However, the same Hammer pattern forming in a High Volatility regime, amidst a panic-driven sell-off, is a much more significant event, signaling a potential capitulation bottom. The primary failure mode in a high-volatility environment shifts from a simple failed reversal to a "stop-hunt," where algorithms deliberately probe below a pattern's low to trigger mass stop-loss orders before the actual bottom is formed [21]. The protocol's emphasis on a 1:2 risk-reward ratio and confirmation filters is an attempt to build a defense against this specific failure mode, but its effectiveness in the post-2020 era remains an empirical question.

The 70/30 data weighting—70% focused on the post-2020 period—is a critical constraint designed to ensure the protocol is validated against contemporary market dynamics [21]. Pre-2020 data is relegated to a secondary role, useful only for verifying the basic geometric integrity of the patterns themselves, not for assessing their predictive power [140]. This is a pragmatic approach, acknowledging that the old rules may no longer apply. For example, a study might show a 60% win rate for Bullish Engulfing patterns on the S&P 500 from 1990-2019. However, if under the same protocol's filter stack, this win rate collapses to 48% in the 2020-2026 period, the pattern would be deemed unsuitable for current market conditions, regardless of its historical performance [78]. This approach forces a hard truth: an edge that cannot survive in the current market regime is not an edge at all. The prevalence of algorithmic liquidity providers has changed the way markets digest information and react to technical levels. Stops are now hunted with precision, and patterns can be manipulated to look like reversals only to continue in the original trend direction. This makes the confirmation filter and realistic slippage modeling absolutely essential for a valid test [27] [95].

The provided source materials, while lacking direct backtest data for the specified protocol, strongly support the premise that market regimes matter profoundly. Research shows that financial time series exhibit regime-switching behavior between states of low-volatility growth and high-volatility turmoil [21]. Other studies have explored the effects of financial risk cycles on business cycles, demonstrating that economies shift between periods of stability and crisis [4] [18]. The VIX itself is shown to be a key indicator of investor expectations and market stress [55] [57]. The term structure of variance risk, which describes how volatility is priced over different time horizons, is also shown to change with market conditions [1]. This body of work collectively validates the analytical framework of stratifying performance by volatility regime. Without this regime-aware analysis, any conclusion about a pattern's win rate would be an unweighted average, masking severe performance degradation in the very environments where traders need a

reliable edge the most. The task of empirically validating the Protocol 3 Tier 2 patterns is therefore not just about finding a number, but about understanding how that number changes depending on the market's state. The ultimate verdict for each pattern will depend on its ability to maintain a filtered win rate above 52% across all relevant post-2020 volatility regimes for each specific instrument [140].

Synthesis and Strategic Recommendations

This research report sought to empirically validate the win rates and net expectancy of Protocol 3 Tier 2 candlestick patterns under a strict 8-filter convergence stack and instrument-specific calibrations. The analysis was guided by a rigorous framework emphasizing post-2020 market conditions, realistic transaction cost modeling, and a non-aggregated, per-instrument evaluation. The investigation reveals a critical gap in the publicly available empirical literature: there are no readily accessible institutional or proprietary backtests that precisely replicate the user-defined protocol. Consequently, definitive, quantitative answers regarding win rates and expectancy for each pattern-instrument combination are not available in the provided sources. The following recommendations are synthesized from the qualitative evidence and analytical reasoning presented throughout this report, serving as a strategic guide for both interpreting the protocol's theoretical standing and for conducting the necessary future validation.

First, the core hypothesis of the research—that a filter-dependent edge exists—is theoretically sound and logically consistent with modern trading principles. The protocol's multi-layered filtering system, incorporating trend alignment, momentum exhaustion, volatility thresholds, and session timing, is designed to systematically eliminate low-probability setups and isolate high-quality signals [87] [140]. The requirement for a 1:2 risk-reward ratio and confirmation-based entries further underscores a professional approach aimed at preserving capital and maximizing gains from winning trades. However, the absence of direct empirical data means this edge remains a hypothesis awaiting rigorous testing. The protocol's design, particularly the instrument-specific calibrations, demonstrates a sophisticated understanding of market microstructure, suggesting it is built on a solid foundation of practical trading experience [26] [140].

Second, the strategic recommendation is to proceed with a new, targeted backtesting initiative. Given the lack of existing data matching the protocol's specificity, this is the only path to achieving the research goal. The backtest must be designed according to the

methodological framework outlined in this report: 1. **Data Acquisition:** Secure granular historical data (tick or 1-minute data preferred) for each instrument class (Forex Majors, XAUUSD, XAGUSD, USOIL, and relevant Index CFDs) covering the 2020-2026 period. 2. **Backtesting Environment:** Utilize a platform capable of modeling realistic execution friction, including variable spreads, slippage models based on ATR or volume, and accounting for session-specific liquidity variations and, where applicable, rollover costs for futures-based CFDs [45] [95]. 3. **Protocol Implementation:** Code the protocol's mechanical definitions and the full 8-filter stack exactly as specified, including all instrument-specific overrides. 4. **Analysis and Reporting:** Execute the backtest and generate performance reports for each pattern/instrument combination. Stratify the results by VIX regime (<16, 16-25, >25) and calculate win rates, net expectancy (in R-multiples), and Sharpe ratios. All results with a sample size below N=30 must be clearly flagged with their 95% confidence intervals [78].

Third, based on the qualitative analysis of each pattern's mechanics and the unique characteristics of each asset class, preliminary verdicts can be tentatively assigned. These verdicts are not final and are intended to highlight areas of potential strength and weakness within the protocol.

- **Keep:** The multi-candle patterns (Morning/Evening Star, Three Inside Up/Down) and the standard single-candle patterns (Hammer, Shooting Star, Engulfing) applied to Forex Majors and Gold are recommended for retention. These assets have relatively stable microstructures where trend-following and reversal patterns are more likely to behave in a predictable manner. The calibrations for Gold are reasonable adaptations to its specific volatility [26].
- **Downgrade:** The same patterns applied to Silver are recommended for a downgrade to a "filter-only" status. Silver's extreme volatility and propensity for whipsaws suggest that its standalone pattern-based signals are unreliable. The patterns may still serve as useful confluence indicators within a broader, more robust strategy, but they lack the standalone reliability required for Tier 2 status. The empirical validation must focus on whether the aggressive calibrations (high ATR filter, wide RSI bands) are sufficient to tame the pattern's inherent noise [140].
- **Conditional Evaluation:** The patterns applied to Oil and Indices require special attention. For Oil, the heavy reliance on news blackouts and the "fundamentals override" rule means that the technical patterns are secondary tools. Their value is contingent on their ability to provide reliable signals in the 70-90 minutes surrounding major data releases. For Indices, the VIX-tiered rules are complex but logical. The key question is whether the protocol can successfully navigate the different behavioral regimes of indices (calm, normal, panicked). The empirical test must determine if the widening of filters and reduction in size at high VIX truly

improves performance or simply reduces the number of trades without improving quality.

In conclusion, the Protocol 3 Tier 2 patterns, as defined, represent a highly structured and theoretically plausible approach to trading. However, their practical viability is unproven by the available evidence. The primary strategic recommendation is to conduct the prescribed backtest to generate the empirical data needed for a definitive verdict. Until then, the patterns should be treated with caution, with particular skepticism reserved for their application in the high-volatility environments characteristic of Silver, Oil, and Indices. The journey from a well-designed protocol to a consistently profitable strategy is paved with empirical validation, and this report identifies the necessary steps to undertake that validation.

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